Peak Prediction Using Quantile Regression and Asymmetric Laplace Noise

Abstract

Predicting the location of peaks in time series is important when decisions must be made before the peak is fully observed. We model the observations as a Gaussian trend with additive noise and develop a quantile-based procedure for estimating its amplitude, location, and spread. For fixed location and spread, the amplitude estimator is a weighted τ-quantile of a transformed version of the observations. We introduce a τ-aligned asymmetric Laplace noise model whose negative log-likelihood is proportional to the quantile check loss, providing a probabilistic interpretation of the criterion and allowing the effects of the quantile level τ and the signal-to-noise ratio to be studied in a unified framework. When the estimated peak location does not stabilize early enough to yield an accurate point prediction, we propose a two-stage construction based on the oscillations of the sequential estimates. It produces candidate entry and exit times, and hence candidate peak regions, using only the estimation trajectory and not the true peak location. The observed peak can be used retrospectively to evaluate the candidates. The methodology is illustrated on daily COVID-19 cases in China, France, and Germany.

Keywords: Peak prediction; quantile regression; asymmetric Laplace noise; peak-region entry and exit times; time series; COVID-19.

Mathematics subject classification: 62M10, 62G35, 60G35.
JEL classification: C22, C21, C53, I18.

1 Introduction

Predicting peaks in time series data is important across a wide range of applications involving proactive decision-making, risk management, resource allocation, and disaster preparedness, such as anticipating disease outbreaks in epidemiology or peak demand in energy management. Peak prediction is challenging because observations are affected by noise, and the underlying signal may exhibit nonlinear or otherwise complex behavior. Measurement errors and random fluctuations can make the location and amplitude of a peak difficult to identify from partial observations, and real-world time series may contain multiple peaks, irregular peak shapes, or other temporal structures that are not adequately described by simple forecasting models.

Popular forecasting approaches such as Autoregressive Integrated Moving Average (ARIMA) models and Recurrent Neural Networks (RNNs) can be effective for describing temporal dependence and forecasting general patterns, but they are not specifically designed to estimate the timing and magnitude of a future peak. A model that provides a good overall forecast may not necessarily provide an accurate estimate of the location of a future peak. There has also been comparatively little theoretical work devoted specifically to peak prediction in time series; one contribution in this direction is the thesis of Stewart et al. (2023), which studies the interplay between trend, seasonality, and noise (see also Yu et al. (2019); Pei and Shaman (2017)).

Our approach builds on the framework developed in Dermoune et al. (2026). We consider observations available at an observation horizon T and a prescribed future horizon $h > T$. The observations are modeled, as in Section 2, as a Gaussian trend of amplitude $A > 0$, peak location L, and spread $S > 0$, corrupted by additive noise. The objective is to estimate (A, L, S) from the observations available up to time T and, in particular, to predict the location L of the peak before it is fully observed.

Our first contribution is to replace the least-absolute-deviation criterion used in the symmetric case by a general quantile regression criterion indexed by $\tau \in (0, 1)$. As we show in Section 2, for fixed location and spread, the amplitude minimizing this criterion is a weighted τ-quantile of a transformed version of the observations; the weighted median of the symmetric case is recovered at $\tau = 1/2$.

Our second contribution is a probabilistic interpretation of the criterion. In Section 4 we introduce a τ-aligned asymmetric Laplace noise model whose negative log-likelihood is proportional to the quantile check loss. This allows the effects of the quantile level τ and of the signal-to-noise ratio to be studied within a unified framework, and it recovers the symmetric Laplace model at $\tau = 1/2$. Sections 5 and 6 examine the resulting estimated peak-location trajectories under high and low signal-to-noise ratios.

Our third contribution concerns the situation in which the estimated peak location does not stabilize early enough to yield an accurate point prediction. In this work, we make precise the construction of the entry and exit times of the peak region, based on the oscillations of the sequential estimates $T \mapsto \hat{l}(T)$. The construction determines candidate entry and exit times, and hence candidate peak regions, using only the estimation trajectory; knowledge of the true peak location is not required operationally, and when it is available it is used only

retrospectively to evaluate the candidates. The methodology is applied to daily COVID-19 case counts during the first wave of the pandemic in China (Barmparis and Tsironis, 2020), France, and Germany, for several quantile levels (Section 8).

The remainder of the paper is organized as follows. In Section 2, we introduce the Gaussian peak model and the quantile-based estimation procedure, including the weighted τ-quantile characterization of the amplitude estimator. Section 3 studies the behavior of the quantile-based objective in the noise-free setting. Section 4 introduces the τ-aligned asymmetric Laplace noise model and establishes its connection with the quantile loss. Section 5 investigates the trajectories of the estimated peak parameters under high signal-to-noise ratio. Section 6 examines the temporal stability of the estimated peak location when the signal-to-noise ratio is reduced. Section 7 makes explicit the construction of the candidate entry and exit times based on the oscillations of the sequential peak-location estimates, and defines the associated candidate peak regions. Finally, Section 8 applies the proposed methodology to daily COVID-19 case data from China, France, and Germany.

2 Quantile-Based Estimation Methodology

We consider the problem of estimating the amplitude, location, and spread of a unimodal signal corrupted by noise. The model is:

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \qquad t = 1, \dots, T. \tag{1}$$

where $A > 0$ is the amplitude, $L > 0$ the peak location, $S > 0$ the spread (scale), and $e(t)$ is additive noise. Given observations $(I(t))_{t=1}^{T}$, our goal is to estimate the underlying parameters using quantile regression, i.e., minimizing the asymmetric loss

$$\rho_\tau(u) = u\left(\tau - \mathbf{1}_{\{u<0\}}\right). \tag{2}$$

for a fixed quantile level $\tau \in (0, 1)$. We define the quantile objective:

$$f_\tau(T, a, l, s) = \frac{1}{T}\sum_{t=1}^{T} \rho_\tau\left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right). \tag{3}$$

For fixed candidate values of (l, s), the optimal amplitude a minimizes $a \mapsto f_\tau(T, a, l, s)$. This minimizer is a weighted τ-quantile of a transformed version of the data, as described below.

2.1 Weighted Quantile Estimation of Amplitude

Let $w(t) = \exp\left(-\frac{(t-l)^2}{s^2}\right)$ be the weight associated with time point t, and define:

$$x(t) = I(t) \exp\left(\frac{(t-l)^2}{s^2}\right).$$

Then the optimal amplitude a is the weighted τ-quantile of $x(t)$ with weights $w(t)$. The following proposition gives the corresponding characterization of the weighted quantile as the minimizer of the weighted quantile loss. The proof of the following Proposition is the same as in the work of Novoselac (2020).

Proposition 1. *Let $(x(t), w(t))$ be a sequence of real values with positive weights $w(t) > 0$. Then, for any $\tau \in (0,1)$, the weighted τ-quantile minimizes the weighted quantile loss:*

$$q_\tau = \arg\min_{a \in \mathbb{R}} \sum_{t=1}^{T} w(t)\, \rho_\tau\big(x(t) - a\big),$$

where ρ_τ is the quantile check function defined in (2). To compute q_τ, sort the values $x(t)$ in increasing order, obtaining a permutation π such that

$$x(\pi(1)) \le x(\pi(2)) \le \cdots \le x(\pi(T)).$$

Define the reordered weights $w_\pi(t) = w(\pi(t))$. Let

$$k = \min\left\{ j \in \{1, \ldots, T\} : \sum_{t=1}^{j} w_\pi(t) \ge \tau \sum_{t=1}^{T} w(t) \right\}.$$

Then the weighted τ-quantile satisfies $q_\tau = x(\pi(k))$.

2.2 Quantile-Based Estimation Procedure

We define the following steps to jointly estimate the location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and spread $\hat{s}(T)$:

Algorithm 1 Quantile-based estimation of peak parameters

Require: Time horizon T, upper bound $h > T$, quantile level $\tau \in (0,1)$

1: For each candidate $l \in [1, h]$, find $s^*(l) = \arg\min_s f_\tau(T, a^*(l,s), l, s)$
2: Set $\hat{l}(T) = \arg\min_l f_\tau(T, a^*(l, s^*(l)), l, s^*(l))$
3: Set $\hat{s}(T) = s^*(\hat{l}(T))$
4: Set $\hat{a}(T) = a^*(\hat{l}(T), \hat{s}(T))$

Ensure: Estimated parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$

2.3 Remark on Asymmetry and Robustness

This method generalizes the classic Least Absolute Deviations (LAD) estimator, recovered at $\tau = 0.5$. By adjusting τ, we introduce asymmetry into the estimation loss, enabling the model to better adapt to skewed noise distributions. This makes the approach especially valuable in early-warning systems and epidemic peak detection where overestimation and underestimation have asymmetric consequences.

3 Quantile-Based Objective Function under Noise-Free Conditions

We first consider the idealized scenario where the signal is not corrupted by noise. The model of interest is given by

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right), \quad t = 1, \ldots, T,$$

where $A = 15000$, $L = 22$, $S = 10$, and $T \in \{2, 3, 10, 20, 21\}$. The quantile-based estimation procedure is implemented by minimizing the following objective function (3). The amplitude a is computed via a weighted τ-quantile, as described in Section 2.

Since the goal is to predict the location L of the peak before it becomes observable, we restrict the minimization over candidate peak locations $l > T$. For each value of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute and visualize the objective curve $l \mapsto f_\tau(T, a(T,l), l, s(T,l))$ for the selected values of T (see Figure 1).

Remark. For each fixed sample size T, we computed the value of the minimized objective function at the estimated parameters $(\hat{a}, \hat{l}, \hat{s})$. The results indicate that the lowest objective value is typically achieved for $\tau = 0.95$, suggesting that upper quantiles align particularly well with the noiseless signal structure. The only exception is at $T = 20$, where $\tau = 0.75$ yields a slightly smaller loss. This empirical observation may help guide quantile selection when prior knowledge of signal shape or noise properties is limited.

4 Quantile-Based Estimation with τ-Aligned Noise

To analyze how quantile-aligned noise affects peak estimation, we introduce a unified noise model based on the *general asymmetric Laplace* (GAL) distribution. This flexible family allows both symmetric and asymmetric noise structures while maintaining the property that the τ-quantile of the noise is at zero.

4.1 Asymmetric Laplace Noise Aligned with the τ-Quantile

We consider an asymmetric Laplace distribution parametrized by a location parameter $m \in \mathbb{R}$, a scale parameter $\lambda > 0$, and an asymmetry parameter $k > 0$. Its density is given by

$$f_{m,\lambda,k}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\left(\dfrac{\lambda(x-m)}{k}\right), & x < m, \\ \exp\left(-\lambda k(x-m)\right), & x \geq m. \end{cases}$$

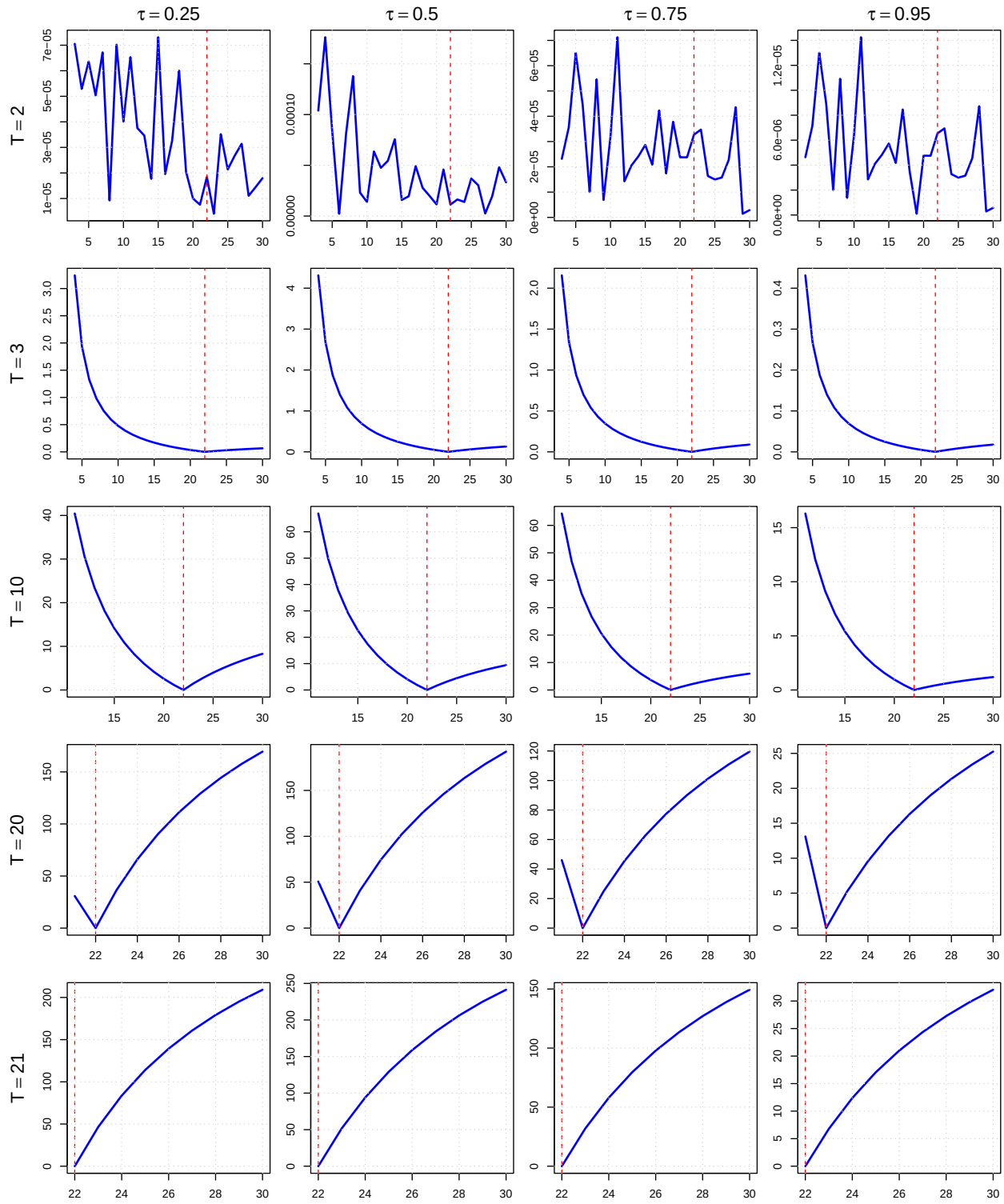


Figure 1: Objective function $l \mapsto f_\tau(T, a, l, s)$ computed for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{2, 3, 10, 20, 21\}$, under a noise-free signal. The vertical dashed line indicates the true peak location $L = 22$, and the search is restricted to values $l > T$.

The parameter m is the mode of the distribution, while k controls its asymmetry and λ controls its concentration. The distribution function evaluated at the mode is

$$\mathbb{P}(X < m) = \frac{\lambda}{k + k^{-1}} \int_{-\infty}^{m} \exp\left(\frac{\lambda(x-m)}{k}\right) dx = \frac{k^2}{1+k^2}.$$

Therefore, to align the asymmetric Laplace noise with the τ-quantile regression criterion, we require

$$\mathbb{P}(e(t) < 0) = \tau, \tag{4}$$

and $\tau \in (0,1)$. Since the location of the noise can be fixed at zero, we set $m = 0$. The corresponding asymmetry parameter is then determined by

$$\frac{k^2}{1+k^2} = \tau,$$

or equivalently,

$$k = \sqrt{\frac{\tau}{1-\tau}}. \tag{5}$$

Thus, the τ-aligned asymmetric Laplace noise has density

$$f_{\tau,\lambda}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\left(\dfrac{\lambda x}{k}\right), & x < 0, \\ \exp(-\lambda k x), & x \geq 0, \end{cases}$$

where k is given by (5). By construction, the noise satisfies the τ-alignment property (4). In particular, when $\tau = 1/2$, we have $k = 1$, and the asymmetric Laplace distribution reduces to the symmetric Laplace distribution centered at zero. Hence, the symmetric Laplace case is recovered as a special case of the τ-aligned asymmetric Laplace model.

4.2 Simulation under Asymmetric Laplace Noise

We now investigate the behavior of the quantile-based peak estimation procedure when the additive noise follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1. In the parameterization based on the quantile loss, the density of the noise can be written as

$$f_{\tau,b}(x) = \frac{\tau(1-\tau)}{b} \exp\left(-\frac{1}{b}\rho_\tau(x)\right), \qquad \tau \in (0,1), \quad b > 0. \tag{6}$$

where ρ_τ is the check function defined in (2). Equivalently,

$$f_{\tau,b}(x) = \begin{cases} \dfrac{\tau(1-\tau)}{b} \exp\left(-\dfrac{\tau x}{b}\right), & x \geq 0, \\ \dfrac{\tau(1-\tau)}{b} \exp\left(\dfrac{(1-\tau)x}{b}\right), & x < 0. \end{cases}$$

The normalization follows from

$$\int_{-\infty}^{+\infty} \exp\left(-\frac{1}{b}\rho_\tau(x)\right) dx = \frac{b}{1-\tau} + \frac{b}{\tau} = \frac{b}{\tau(1-\tau)}.$$

Moreover, by construction, the noise satisfies the τ-alignment property (4). This distribution is directly compatible with the quantile-based estimation criterion. Indeed,

$$-\log f_{\tau,b}(e) = -\log\left(\frac{\tau(1-\tau)}{b}\right) + \frac{1}{b}\rho_\tau(e).$$

Therefore, for fixed τ and b, minimizing the negative log-likelihood is equivalent to minimizing the empirical quantile loss. The asymmetric Laplace model thus provides a probabilistic interpretation of the quantile-based peak estimation procedure without introducing a separate estimation criterion. For the simulation study, the true signal is

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right),$$

with $A = 1$, $L = 22$, $S = 10$. Since the maximum amplitude of the Gaussian signal is $\max_t I(t) = A$, we define the signal-to-noise ratio as the ratio between the peak amplitude of the signal and the standard deviation of the noise:

$$\mathrm{SNR} = \frac{A}{\sqrt{\mathrm{Var}(e(t))}}. \tag{7}$$

In the simulations, we set SNR = 1000. For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, the corresponding asymmetric Laplace noise is generated with asymmetry parameter (5). Thus, the noise variance is calibrated according to

$$\mathrm{Var}(e(t)) = \left(\frac{A}{\mathrm{SNR}}\right)^2. \tag{8}$$

For the τ-aligned asymmetric Laplace distribution, this variance is

$$\mathrm{Var}(e(t)) = b^2 \frac{\tau^2 + (1-\tau)^2}{\tau^2(1-\tau)^2}. \tag{9}$$

Combining (8) and (9) gives

$$b = \frac{A}{\mathrm{SNR}} \frac{\tau(1-\tau)}{\sqrt{\tau^2 + (1-\tau)^2}}. \tag{10}$$

For each τ, the objective function $l \mapsto f_\tau(T, a, l, s)$ is evaluated for the time horizons $T \in \{4, 6, 8, 9, 10\}$. For each candidate peak location l, the scale parameter s is optimized over the prescribed interval, and the amplitude a is determined as the weighted τ-quantile described in Section 2.1. The resulting objective profiles are then compared across the different values of τ. The resulting profiles are displayed in Figure 2. The true peak location $L = 22$ is marked by a vertical dashed line. Each panel corresponds to a different pair (T, τ), and the curves show the behavior of the quantile-loss objective across candidate peak locations.

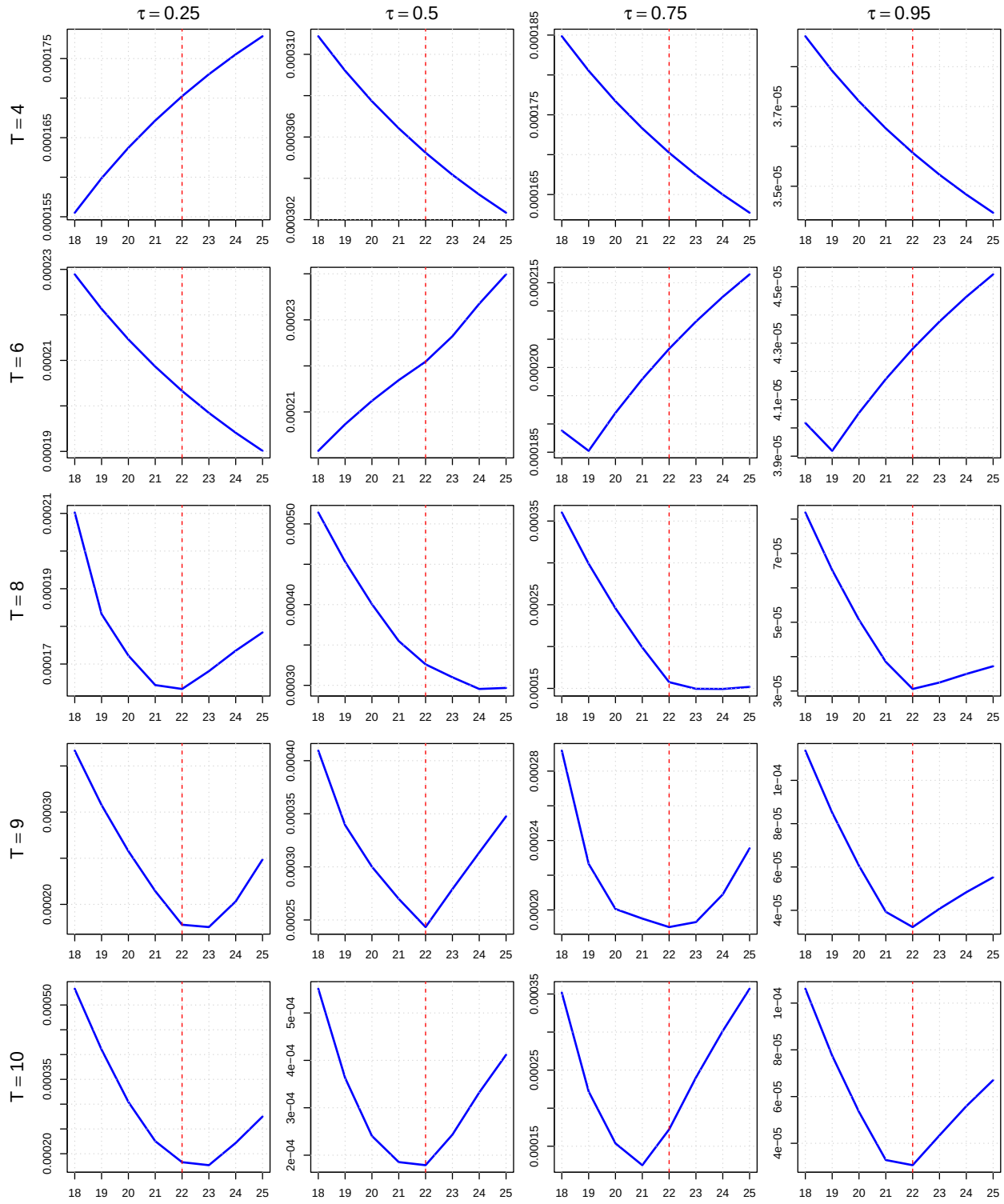


Figure 2: Objective function profiles $l \mapsto f_\tau(T, a, l, s)$ under *asymmetric Laplace* noise for all combinations of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ (columns) and $T \in \{4, 6, 8, 9, 10\}$ (rows). Each subfigure shows the loss function for a specific T and τ pair, with the dashed vertical line indicating the true peak location $L = 22$.

5 Parameter Estimation Trajectories

We now analyze how the estimated model parameters, location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and scale $\hat{s}(T)$, evolve as the sample size T increases. Our goal is to assess the stability and convergence of these estimators under the τ-aligned asymmetric Laplace noise introduced in Section 4.1, including its symmetric special case. All experiments use the SNR definition given in (7), with SNR $= 1000$, and consider quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$.

5.1 τ-Aligned Asymmetric Laplace Noise

We consider the signal-plus-noise model (1), with $h = 30$, $A = 1$, $L = 22$, $S = 10$. The noise $e(t)$ follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1, with density (6). The asymmetry parameter is determined by the quantile level according to (5), so that the τ-alignment property (4) holds. In particular, $\tau = 1/2$ corresponds to the symmetric Laplace distribution, whereas the other values of τ considered here produce asymmetric noise. The scale parameter b is calibrated separately for each τ so that the variance of the noise satisfies the fixed signal-to-noise ratio condition (8). For the τ-aligned asymmetric Laplace distribution, this gives (9), and therefore (10). At each time horizon $T \in \{4, \ldots, 30\}$, we estimate the parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$ by minimizing the objective function (3) over the candidate peak locations and scale parameters, with the amplitude a re-estimated for each pair (l, s) as the weighted τ-quantile described in Section 2.1. The scale parameter s is optimized over the prescribed interval, and the location parameter l is selected by minimizing the resulting objective over the candidate locations. Figure 3 shows the resulting estimation trajectories $T \mapsto \hat{l}(T)$, $T \mapsto \hat{a}(T)$, $T \mapsto \hat{s}(T)$, for each quantile level τ. The true parameter values $(L, A, S) = (22, 1, 10)$ are indicated by dashed horizontal lines.

6 Stability Analysis

This section investigates the temporal stability of the estimated peak location when the signal-to-noise ratio is reduced. We focus on the τ-aligned asymmetric Laplace noise introduced in Section 4.1 and examine how the estimated peak location evolves as new observations become available. In particular, we compare the stability of the estimation paths across different quantile levels and assess the effect of a lower signal-to-noise ratio on the ability to recover the true peak location.

6.1 Low Signal-to-Noise Performance

To evaluate the robustness of the quantile-based estimator under a lower signal-to-noise ratio, we consider the signal-plus-noise model (1), with $h = 35$, $A = 1$, $L = 22$, $S = 10$. The noise $e(t)$ follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1, with density (6). For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, the scale parameter b is calibrated so

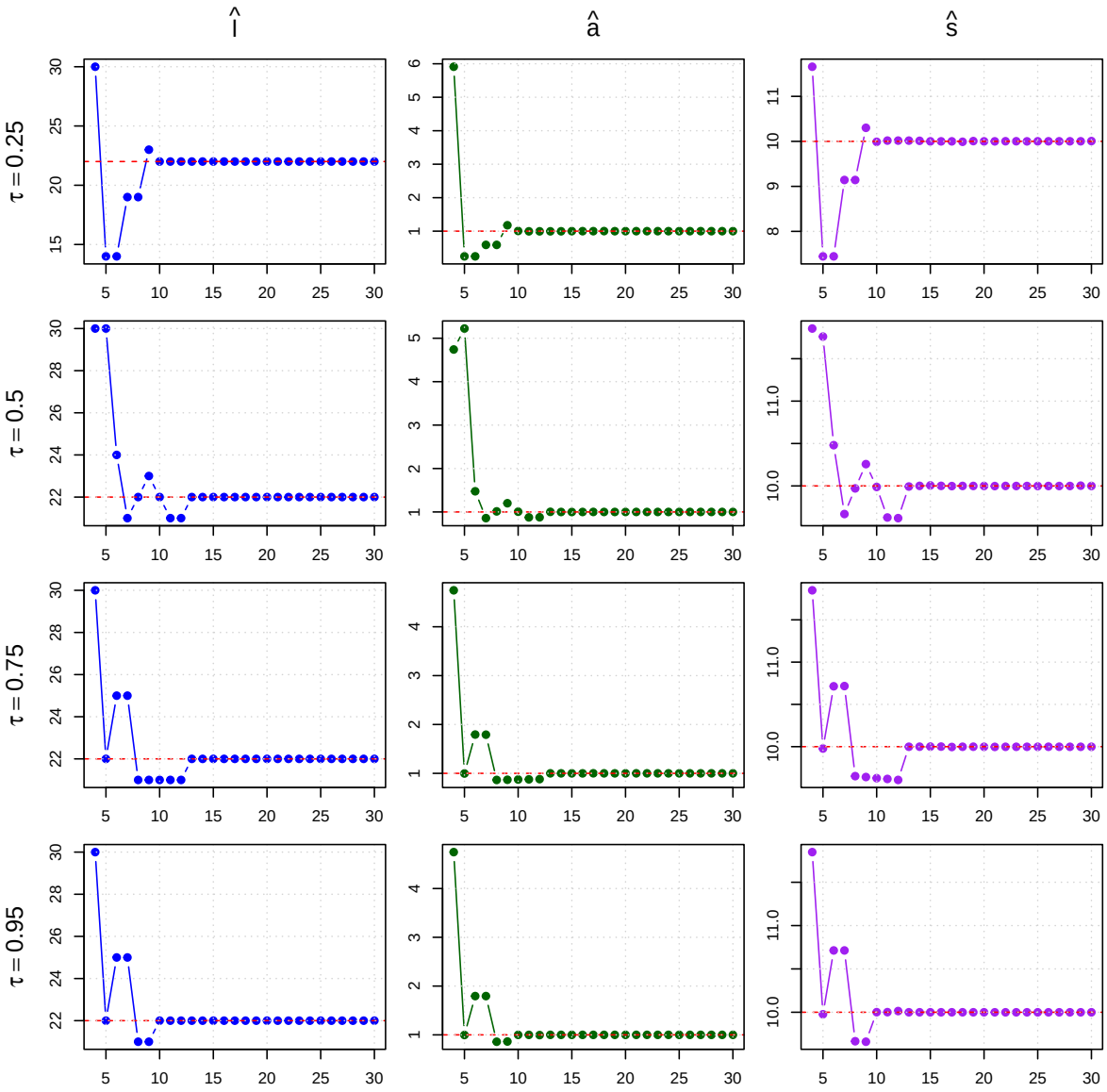


Figure 3: Estimation paths $T \mapsto \hat{l}(T)$ (left), $\hat{a}(T)$ (middle), and $\hat{s}(T)$ (right) under τ-aligned asymmetric Laplace noise with signal-to-noise ratio (SNR) 1000. Each row corresponds to a quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The true values $(L, A, S) = (22, 1, 10)$ are indicated by dashed horizontal lines.

that (8), with SNR = 40, according to the definition in (7). Using the variance of the τ-aligned asymmetric Laplace distribution gives (10). For each quantile level τ, we compute the estimation path $T \mapsto \hat{l}(T)$, $T = 4, \ldots, 35$, where $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$ is obtained by minimizing the empirical quantile loss over the candidate peak locations and scale parameters, with the amplitude re-estimated as the weighted τ-quantile described in Section 2.1. To characterize the temporal stability of the estimated peak location, we consider the earliest observation horizon T_{stab} from which all subsequent estimates remain within a prescribed range of one another. Specifically, for a tolerance $\varepsilon = 4$, we define

$$T_{\text{stab}} = \min \left\{ T : \max_{u \geq T} \hat{l}(u) - \min_{u \geq T} \hat{l}(u) \leq \varepsilon \right\}.$$

This stabilization time measures the earliest point from which the remaining estimation path remains confined to a stable range. Figure 4 displays the resulting estimation paths for the four quantile levels. The horizontal dashed line indicates the true peak location $L = 22$. The comparison illustrates how the temporal stability of the estimated peak location depends on the quantile level when the signal-to-noise ratio is reduced. The corresponding stabilization times are reported in Table 1. Smaller values of T_{stab} indicate that the estimator reaches a stable regime earlier, whereas larger values indicate that additional observations are required before the estimated peak location becomes stable.

Table 1: Stabilization times T_{stab} for the estimated peak location under SNR = 40 and different quantile levels.

	$\tau = 0.25$	$\tau = 0.5$	$\tau = 0.75$	$\tau = 0.95$
T_{stab}	17	17	11	10

7 Entry and Exit Times of the Peak Region

The sequential estimation procedure introduced in the preceding sections provides, for each observation horizon T, an estimate $\hat{l}(T)$ of the peak location. Here, T denotes the current observation horizon, so that the estimator is constructed using only the observations available on $[1, T]$, while the candidate peak location is searched over the fixed horizon $[1, h]$, with $T < h$ and $L < h$. Thus, although only the observations $1, \ldots, T$ are used in the estimation, the estimator may identify a peak location beyond the current observation horizon T. For notational simplicity, we write $\hat{l}(T)$ instead of $\hat{l}(T, h)$ when the horizon h is fixed.

The stabilization criterion considered in Section 6 characterizes the point from which the future estimation path remains within a prescribed tolerance. In this section, we introduce a complementary construction based on the oscillations of the estimated peak location. This construction aims at identifying candidate time intervals describing the region in which the estimated peak location becomes stable, through a data-driven time of entry and a time of exit.

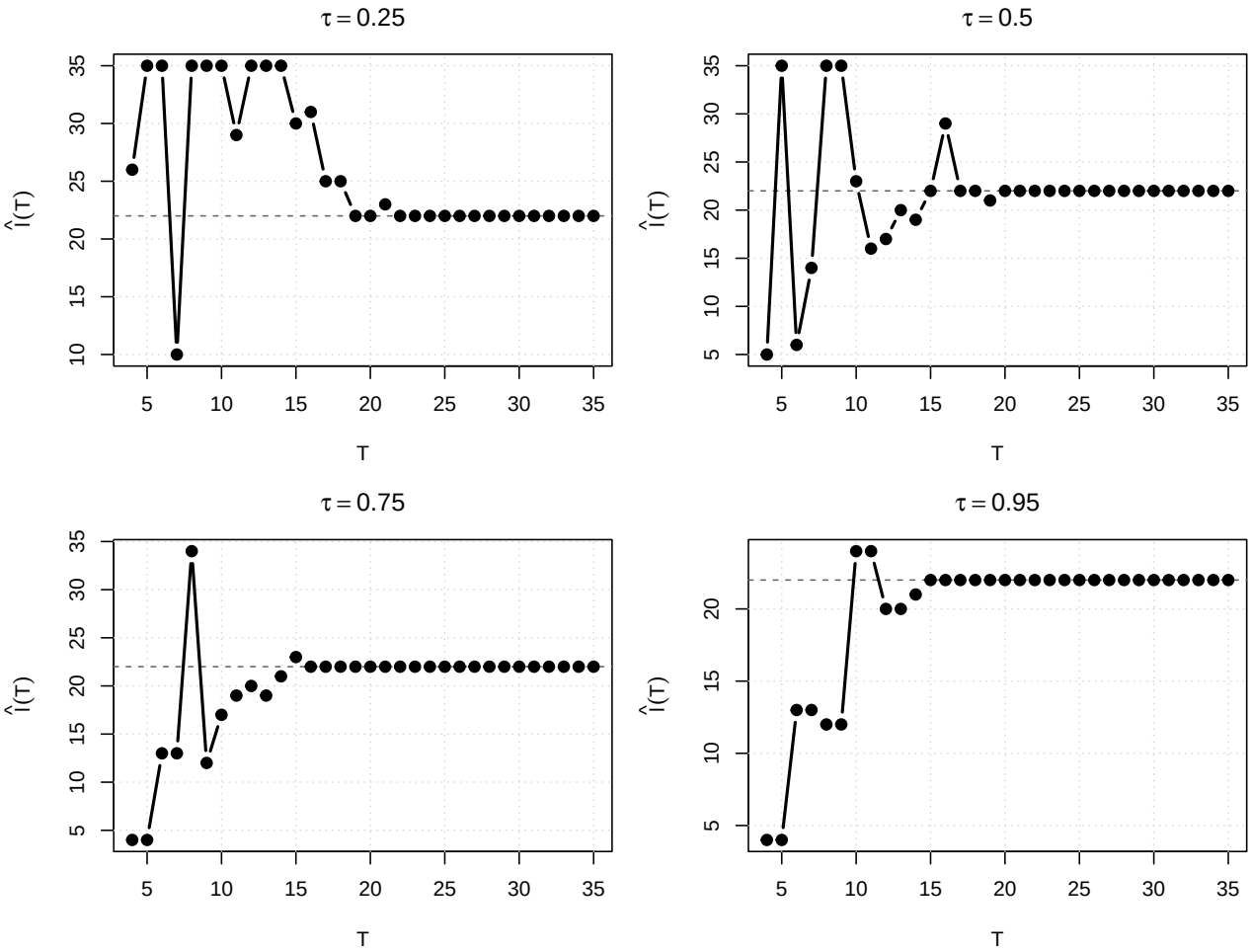


Figure 4: Estimation paths $T \mapsto \hat{l}(T)$ under a low signal-to-noise ratio (SNR $= 40$), for quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The horizontal dashed line indicates the true peak location $L = 22$.

The operational construction is based only on the sequential estimation trajectory $T \longmapsto \hat{l}(T)$. The true peak location L is not required to construct the candidate entry and exit times. In the simulation study, where L is known, it is used only for an ex post assessment of whether a candidate peak region contains the true peak.

7.1 Oscillations of the estimated peak location

We fix a sufficiently large horizon h such that the true peak location satisfies $L < h$. For each observation horizon $T < h$, the sequential estimator $\hat{l}(T)$ is constructed using only the observations available on $[1, T]$, while the candidate peak location is searched over $[1, h]$. To characterize the temporal stability of the estimated peak location using only the information available at the current observation horizon T, we consider the oscillations of the estimation path over intervals contained in the observed time range $[1, T]$. For two times t and T satisfying $T_0 \leq t \leq T < h$, we define the oscillation of the estimated peak location over the interval $[t, T]$ by

$$\omega_l(t, T) = \max_{t \leq t_1, t_2 \leq T} \left| \hat{l}(t_1) - \hat{l}(t_2) \right|. \tag{11}$$

Thus, $\omega_l(t, T)$ measures the largest variation of the estimated peak location over the observed interval $[t, T]$. In particular, $\omega_l(T, T) = 0$. For a fixed observation horizon T, the function $t \longmapsto \omega_l(t, T)$, with $T_0 \leq t \leq T$, is nonincreasing. Indeed, if $t_1 < t_2 \leq T$, then $[t_2, T] \subset [t_1, T]$, and consequently $\omega_l(t_2, T) \leq \omega_l(t_1, T)$. The strict decreases of this sequence identify the times at which the oscillation level changes when the starting point of the observed interval is moved forward.

7.2 Strict-decrease points of the retrospective oscillation

For a fixed observation horizon $T < h$, consider the nonincreasing sequence $t \longmapsto \omega_l(t, T)$, with $t \in \{T_0, \ldots, T\}$. We call an observation time $t < T$ a *strict-decrease point* of the retrospective oscillation if

$$\omega_l(t, T) > \omega_l(t + 1, T).$$

We order the strict-decrease points in decreasing order. Thus, if $t_3 < t_2 < t_1 < T$ are the strict-decrease points, then $t_1 > t_2 > t_3$. The intervals between consecutive strict-decrease points correspond to constant plateaus of the nonincreasing sequence $t \mapsto \omega_l(t, T)$. More precisely, if $U > V$ are two consecutive strict-decrease points, then

$$\omega_l(t, T) = \omega_l(U, T), \qquad t \in \{V + 1, \ldots, U\}.$$

Thus, for each current observation horizon T, the strict-decrease points characterize the successive decreases of the retrospective oscillation computed using only the observations available on $[1, T]$.

7.3 Construction of the candidate exit times

For each current observation horizon $T < h$, the candidate exit times are defined from the strict-decrease points of the sequence $t \longmapsto \omega_l(t, T)$. We set $T^{(0)}_{\text{exit}}(T) = T$, and let $T^{(1)}_{\text{exit}}(T) > T^{(2)}_{\text{exit}}(T) > \cdots$ denote the strict-decrease points of $t \mapsto \omega_l(t, T)$ that are strictly smaller than T, ordered in decreasing order. For example, if the strict-decrease points are $t_3 < t_2 < t_1 < T$, then

$$T^{(1)}_{\text{exit}}(T) = t_1, \qquad T^{(2)}_{\text{exit}}(T) = t_2, \qquad T^{(3)}_{\text{exit}}(T) = t_3.$$

The convention $T^{(0)}_{\text{exit}}(T) = T$ provides the initial candidate exit time, for which $\omega_l\left(T^{(0)}_{\text{exit}}(T), T\right) = \omega_l(T, T) = 0$. For each candidate index $k \geq 0$, we define the corresponding oscillation level by

$$\epsilon'_k(T) = \omega_l\left(T^{(k)}_{\text{exit}}(T), T\right).$$

Thus, for each current observation horizon T, the candidate exit times correspond to the successive strict decreases of the retrospective oscillation, while the associated levels $\epsilon'_k(T)$ record the oscillation at those candidate exit times. This construction is based only on the sequential estimation trajectory available up to the current observation horizon T and does not require knowledge of the true peak location L. Operationally, for each fixed T, the candidate exit times are obtained by scanning the sequence $t \longmapsto \omega_l(t, T)$ for its successive strict-decrease points. In numerical implementation, a small tolerance can be used when determining strict decreases.

7.4 Determination of the entry time

For a given current observation horizon $T < h$ and a candidate exit time $T^{(k)}_{\text{exit}}(T)$, we first consider all observation times strictly before the candidate exit time. The set

$$E_k(T) = \left\{ t < T^{(k)}_{\text{exit}}(T) : \omega_l\left(t, T^{(k)}_{\text{exit}}(T)\right) \geq \epsilon'_k(T) \right\}, \tag{12}$$

defines the Stage-2 set. If $E_k(T)$ is empty, the candidate does not yield a valid entry time. Otherwise, we define $T^*_k(T) = \max E_k(T)$. The corresponding oscillation level is $\epsilon_k(T) = \omega_l\left(T^*_k(T), T^{(k)}_{\text{exit}}(T)\right)$. We then consider all observation times $t < T^{(k)}_{\text{exit}}(T)$ at which the oscillation equals $\epsilon_k(T)$. The entry time is defined as the smallest such observation horizon:

$$T^{(k)}_{\text{entry}}(T) = \min\left\{ t < T^{(k)}_{\text{exit}}(T) : \omega_l\left(t, T^{(k)}_{\text{exit}}(T)\right) = \epsilon_k(T) \right\}, \tag{13}$$

provided that such an observation time exists. The width associated with the candidate is

$$w_k(T) = T^{(k)}_{\text{exit}}(T) - T^{(k)}_{\text{entry}}(T). \tag{14}$$

The resulting candidate peak region at observation horizon T is therefore represented by the interval

$$\mathcal{R}_k(T) = \left[T^{(k)}_{\text{entry}}(T), T^{(k)}_{\text{exit}}(T) \right].$$

All endpoints are determined entirely from the sequential estimation trajectory available up to the current observation horizon T and from the corresponding retrospective oscillation construction.

7.5 Construction of a data-driven interval for the peak location

We illustrate the construction of the entry–exit intervals introduced in Sections 7.1–7.4 and describe how one of these intervals can be selected using only the observations available at the current horizon. The simulation setting is the one considered in Sections 4 and 5, with $A = 1, L = 22, S = 10, h = 30$, and quantile levels $\tau \in \{0.25, 0.50, 0.75, 0.95\}$. We consider three signal-to-noise ratios, SNR $\in \{1000, 100, 1\}$, corresponding to large, moderate, and small signal-to-noise regimes, respectively. For each current observation horizon T, the two-stage procedure produces a collection of candidate intervals

$$\mathcal{R}_k(T) = \left[T^{(k)}_{\text{entry}}(T), T^{(k)}_{\text{exit}}(T)\right],$$

where k indexes the candidate exit times constructed in Section 7.3. The width of candidate k is

$$w_k(T) = T^{(k)}_{\text{exit}}(T) - T^{(k)}_{\text{entry}}(T).$$

Our objective is to define an operational selection rule that does not require knowledge of the true peak location L.

Stabilization of the sequential estimator. The relevance of the candidate intervals depends on the behaviour of the sequential estimator $\widehat{l}(T)$. We therefore first examine, by simulation, whether $\widehat{l}(T)$ stabilizes at the true peak location L before the terminal horizon. For each pair (τ, SNR), we generate $M = 200$ independent trajectories and record: (i) the entry time $T^{(0)}_{\text{entry}}(h)$ at the terminal horizon; (ii) the oscillation of $\widehat{l}(T)$ over the last ten horizons $T \in [20, 30]$; (iii) the maximal post-entry error $\max_{T \geq T^{(0)}_{\text{entry}}(h)} |\widehat{l}(T) - L|$; (iv) the proportion of replicates with $\widehat{l}(h) = L$; (v) the proportion of replicates with zero oscillation over $[20, 30]$. Table 2 reports the medians and proportions.

Table 2: Stabilization of the sequential estimator $\widehat{l}(T)$, based on $M = 200$ independent replications per (τ, SNR). For each signal-to-noise ratio, the table reports the median entry time at the terminal horizon, the median oscillation over $T \in [20, 30]$, the median post-entry error, the proportion of replicates with $\widehat{l}(h) = L$, and the proportion with zero oscillation over $[20, 30]$. Ranges indicate variation across the four quantile levels.

SNR	$T^{(0)}_{\text{entry}}(h)$	osc. over $[20, 30]$	post-entry error	prop. $\widehat{l}(h) = L$	prop. stable
1000	10–12	0	0	1.00	1.00
100	16–18	0	0	1.00	0.89–1.00
1	27–28	6–11	1.0–2.5	0.14–0.35	0.00–0.03

Figure 5 displays, for $\tau = 0.50$, twenty trajectories of $\widehat{l}(T)$ at each of the three signal-to-noise

ratios. At SNR = 1000 and SNR = 100, the trajectories converge to $L = 22$ well before the terminal horizon and remain there. At SNR = 1, the trajectories continue to fluctuate over the entire observation window and do not stabilize.

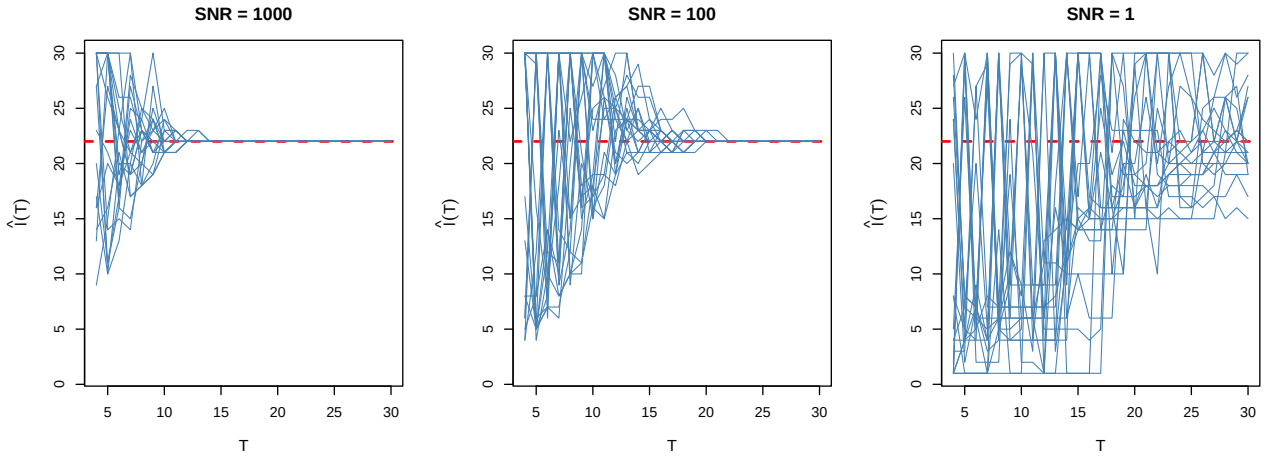


Figure 5: Trajectories $T \mapsto \widehat{l}(T)$ for $\tau = 0.50$, one panel per signal-to-noise ratio, with $M = 20$ independent replicates per panel. The red dashed line indicates the true peak location $L = 22$. At SNR = 1000 and 100, the estimator stabilizes at L; at SNR = 1, it does not.

The distinction has a direct consequence for the analysis that follows. At SNR = 1000 and SNR = 100, the point estimate $\widehat{l}(T)$ alone identifies the true peak location after a short transient, and the entry time $T^{(0)}_{\text{entry}}(T)$ is sufficient. The exit time carries no additional information in these regimes. At SNR = 1, on the other hand, no finite entry time stabilizes the estimator, and the exit times and the admissible set $\mathcal{K}(T)$ become the only tools that bound the peak location. The construction of the candidate intervals, the selection rules, and the empirical coverage study therefore focus on the small-SNR regime, while the high- and medium-SNR cases are reported as boundary cases.

Structure of the candidate intervals. Before defining the selection rule, we record two properties of the candidate family. First, the exit times satisfy

$$T^{(k+1)}_{\text{exit}}(T) < T^{(k)}_{\text{exit}}(T), \qquad k \geq 0, \tag{15}$$

whenever consecutive candidates exist. This follows from the construction, since the nontrivial exit times are the strict-decrease points of $t \mapsto \omega_l(t, T)$, ordered from largest to smallest. Second, the entry times satisfy

$$T^{(k+1)}_{\text{entry}}(T) \leq T^{(k)}_{\text{entry}}(T), \qquad k \geq 0, \tag{16}$$

for all consecutive candidate pairs observed in the simulation at SNR = 1000. Thus, in these simulations, both endpoints move weakly to the left as k increases. These orderings do not imply that the candidate intervals are nested. In particular, the inclusion $\mathcal{R}_{k+1}(T) \subseteq \mathcal{R}_k(T)$ would require $T^{(k+1)}_{\text{entry}}(T) \geq T^{(k)}_{\text{entry}}(T)$, in addition to the exit-time ordering in (15). Moreover,

the entry time of candidate k does not necessarily coincide with the exit time of candidate $k+1$. The exit time is determined from the oscillation $\omega_l(t, T)$ with right endpoint T, whereas the entry time is determined from an oscillation whose right endpoint is $T_{\text{exit}}^{(k)}(T)$. Figure 6 displays the trajectories $T \mapsto T_{\text{exit}}^{(k)}(T)$ and $T \mapsto T_{\text{entry}}^{(k)}(T)$ for $k \in \{0, 1, 2, 3\}$ at SNR $= 1000$, with one panel per quantile level. These plots illustrate the numerical behaviour of the candidate endpoints in the considered setting. The operational selection rule below does not require the candidate intervals to be nested or their consecutive endpoints to coincide.

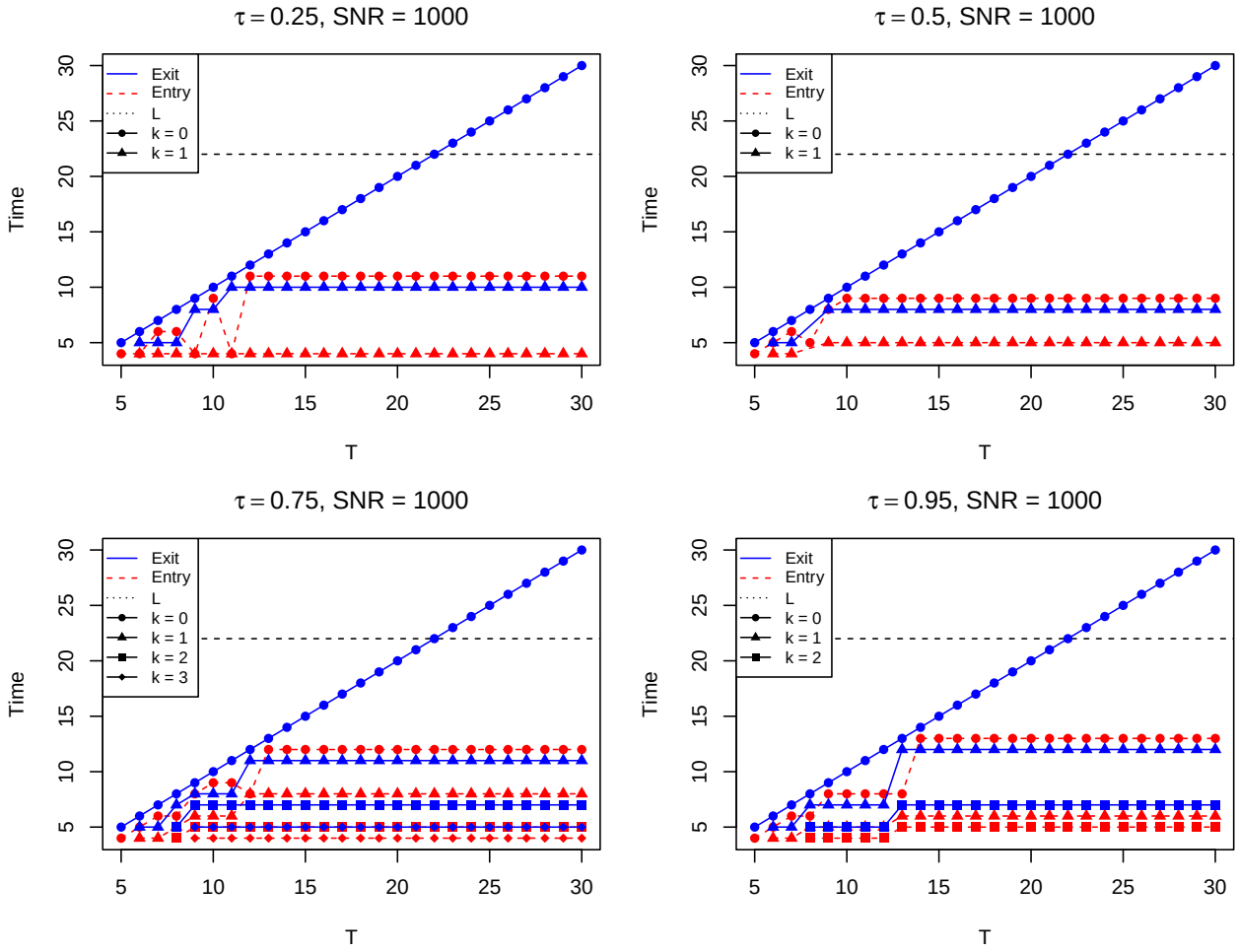


Figure 6: Candidate exit times $T_{\text{exit}}^{(k)}(T)$ (solid curves) and entry times $T_{\text{entry}}^{(k)}(T)$ (dashed curves) as functions of the current horizon T at SNR $= 1000$, for the four quantile levels. The horizontal line indicates the true peak location $L = 22$, shown for simulation reference only.

Operational selection of the interval. At a current horizon T, the sequential estimator $\widehat{l}(T)$ is available, whereas the true peak location L is unknown. We therefore select an interval by requiring it to contain the current point estimate. Define the set of admissible candidate indices by

$$\mathcal{K}(T) = \left\{ k : T_{\text{entry}}^{(k)}(T) \le \widehat{l}(T) \le T_{\text{exit}}^{(k)}(T) \right\}. \tag{17}$$

If $\mathcal{K}(T)$ is nonempty, we select the narrowest admissible candidate:

$$\widehat{k}(T) = \arg\min_{k \in \mathcal{K}(T)} w_k(T), \tag{18}$$

with ties resolved by choosing the smaller k, then the smaller entry time, and finally the smaller exit time. The resulting data-driven interval is

$$\widehat{\mathcal{I}}(T) = \left[T_{\text{entry}}^{(\widehat{k}(T))}(T), T_{\text{exit}}^{(\widehat{k}(T))}(T) \right]. \tag{19}$$

If $\mathcal{K}(T)$ is empty, no interval is selected at that horizon, and the procedure can continue as additional observations become available. We define the first operational selection horizon by

$$T_{\text{est}}(\tau, \text{SNR}) = \min \{ T : \mathcal{K}(T) \neq \varnothing \}. \tag{20}$$

Thus, T_{est} is the earliest observation horizon at which the candidate construction produces an interval consistent with the current estimate. Both the interval and this stopping horizon are computed without using L. In contrast, a criterion based on whether the selected interval contains the true L cannot be used operationally and is therefore not used in the selection rule.

Numerical results. Table 3 reports the first operational selection horizon and the corresponding selected interval for each combination of τ and SNR. The interval width is $w_{\widehat{k}(T_{\text{est}})}(T_{\text{est}})$.

Table 3: First operationally selected candidate interval for each quantile level τ and signal-to-noise ratio. For compactness, write $T^\star = T_{\text{est}}$ and $k^\star = \widehat{k}(T^\star)$. The interval is selected by requiring it to contain $\widehat{l}(T^\star)$ and then minimizing its width over all admissible candidates. The last column reports $w_{k^\star}(T^\star) = T_{\text{exit}}^{(k^\star)}(T^\star) - T_{\text{entry}}^{(k^\star)}(T^\star)$.

τ	SNR	$T^\star$	$k^\star$	$\widehat{l}(T^\star)$	$T_{\text{entry}}^{(k^\star)}(T^\star)$	$T_{\text{exit}}^{(k^\star)}(T^\star)$	$w_{k^\star}(T^\star)$
0.25	1000	22	0	22	11	22	11
0.25	100	22	0	22	18	22	4
0.25	1	6	0	4	4	6	2
0.50	1000	22	0	22	9	22	13
0.50	100	22	0	22	18	22	4
0.50	1	10	1	8	8	9	1
0.75	1000	22	0	22	12	22	10
0.75	100	22	0	22	16	22	6
0.75	1	18	0	17	17	18	1
0.95	1000	22	0	22	13	22	9
0.95	100	22	0	22	16	22	6
0.95	1	9	0	8	8	9	1

For SNR $= 100$ and SNR $= 1000$, the first selected interval occurs at $T_{\text{est}} = 22$ for all four quantile levels. For SNR $= 1$, the first selected horizon varies with τ, taking the values 6, 10, 18, and 9 for $\tau = 0.25$, 0.50, 0.75, and 0.95, respectively. The selected intervals are also substantially

narrower in these low-SNR examples. However, an early and narrow interval need not be close to the true peak location: in the present realization, the low-SNR intervals selected at T_{est} do not contain $L = 22$. This illustrates that the operational criterion guarantees consistency with $\widehat{l}(T)$, not containment of the unknown true peak. Accordingly, $\widehat{\mathcal{I}}(T)$ is a data-driven interval estimate for the peak location. Its construction alone does not establish a confidence level or guarantee

$$\mathbb{P}\Big\{L \in \widehat{\mathcal{I}}(T)\Big\} \geq 1 - \alpha.$$

To interpret this interval as a confidence interval with nominal coverage $1 - \alpha$, an appropriate coverage result must be established theoretically or assessed through repeated independent simulation replicates. The single-realization illustration above is intended to demonstrate the operational construction and its behaviour across quantile levels and signal-to-noise regimes, rather than to establish statistical coverage.

Monte Carlo assessment of empirical coverage. We next assess the empirical coverage of the operational intervals using repeated independent simulations. For each combination of $\tau \in \{0.25, 0.50, 0.75, 0.95\}$ and SNR $\in \{1000, 100, 1\}$, we generate $M = 1000$ independent trajectories under the same peak model, with $A = 1$, $L = 22$, $S = 10$, and observation horizon $h = 30$. At each available horizon T, the operational selection rule defined in (17)–(19) is applied separately to each trajectory. The true peak location L is used only after selection to assess whether the selected interval contains it. Let $\widehat{\mathcal{I}}_m(T)$ denote the selected interval for replicate m, when an interval is available. We define the interval-selection rate by

$$\widehat{q}_M(T) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\Big\{\widehat{\mathcal{I}}_m(T) \neq \varnothing\Big\},$$

and the empirical coverage among selected intervals by

$$\widehat{p}_M^{\text{sel}}(T) = \frac{\displaystyle\sum_{m=1}^{M} \mathbf{1}\Big\{L \in \widehat{\mathcal{I}}_m(T)\Big\}}{\displaystyle\sum_{m=1}^{M} \mathbf{1}\Big\{\widehat{\mathcal{I}}_m(T) \neq \varnothing\Big\}},$$

provided that at least one interval is selected. We also report the unconditional success proportion

$$\widehat{p}_M^{\text{all}}(T) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\Big\{L \in \widehat{\mathcal{I}}_m(T)\Big\}.$$

Thus, $\widehat{p}_{\text{all}}(T)$ accounts for both interval-selection failures and selected intervals that do not contain the true peak location. When no interval is selected in any replicate, the conditional coverage is undefined.

Figure 7 displays the empirical coverage among selected intervals as a function of T, while Figure 8 shows the corresponding interval-selection rates. The results reveal a marked depen-

dence on the signal-to-noise ratio. For SNR = 1000, no interval is selected in the simulations for $T < 22$, whereas the selection rate and empirical coverage are both equal to one from $T = 22$ onward. For SNR = 100, selection is rare at several early horizons, but from $T = 23$ onward all simulated trajectories yield a selected interval, with empirical coverage equal to one. At $T = 22$, the selection rate is between 0.998 and 0.999, and all selected intervals contain L.

The low-SNR results are substantially different. At SNR = 1, intervals may be selected before $T = 22$, but none of the selected intervals at these earlier horizons contains the true peak location in the simulated sample. At $T = 22$, empirical coverage among selected intervals ranges from approximately 0.261 to 0.378, depending on τ. Coverage subsequently increases with the observation horizon, although the rate of increase differs across quantile levels. At $T = 30$, for example, the empirical coverage is approximately 0.613, 0.565, 0.637, and 0.844 for $\tau = 0.25$, 0.50, 0.75, and 0.95, respectively. At this horizon, the interval-selection rate is at least 0.996 for all four quantile levels. Hence, the relatively low coverage at small SNR cannot be attributed solely to failures to select an interval.

Table 4 summarizes the selection rates, empirical coverage probabilities, and mean widths at $T = 30$. In the high- and medium-SNR settings, all selected intervals contain the true peak location in the simulated replicates. Under low SNR, the empirical coverage is lower, despite the high selection rates. The mean widths also differ across the SNR regimes, illustrating that interval width alone does not determine whether an interval contains the true peak.

These findings provide a simulation-based assessment of the operational procedure under the specified model and parameter settings. In particular, they show that an interval can be selected consistently without necessarily achieving high empirical coverage. The results should therefore be interpreted as empirical coverage probabilities under the considered simulation design, rather than as a general theoretical coverage guarantee.

Table 4: Monte Carlo interval-selection rates, empirical coverage, and mean interval widths at $T = 30$, based on $M = 1000$ independent replicates. The conditional coverage is calculated among selected intervals, and the unconditional success proportion also accounts for replicates in which no interval is selected.

τ	SNR	$\widehat{q}_M(30)$	$\widehat{p}_M^{\mathrm{sel}}(30)$	$\widehat{p}_M^{\mathrm{all}}(30)$	Mean width
0.25	1000	1.000	1.000	1.000	18.699
0.25	100	1.000	1.000	1.000	12.663
0.25	1	0.998	0.613	0.612	5.624
0.50	1000	1.000	1.000	1.000	18.259
0.50	100	1.000	1.000	1.000	11.978
0.50	1	0.996	0.565	0.563	5.486
0.75	1000	1.000	1.000	1.000	18.787
0.75	100	1.000	1.000	1.000	12.721
0.75	1	0.996	0.637	0.634	5.296
0.95	1000	1.000	1.000	1.000	19.656
0.95	100	1.000	1.000	1.000	14.200
0.95	1	1.000	0.844	0.844	5.753

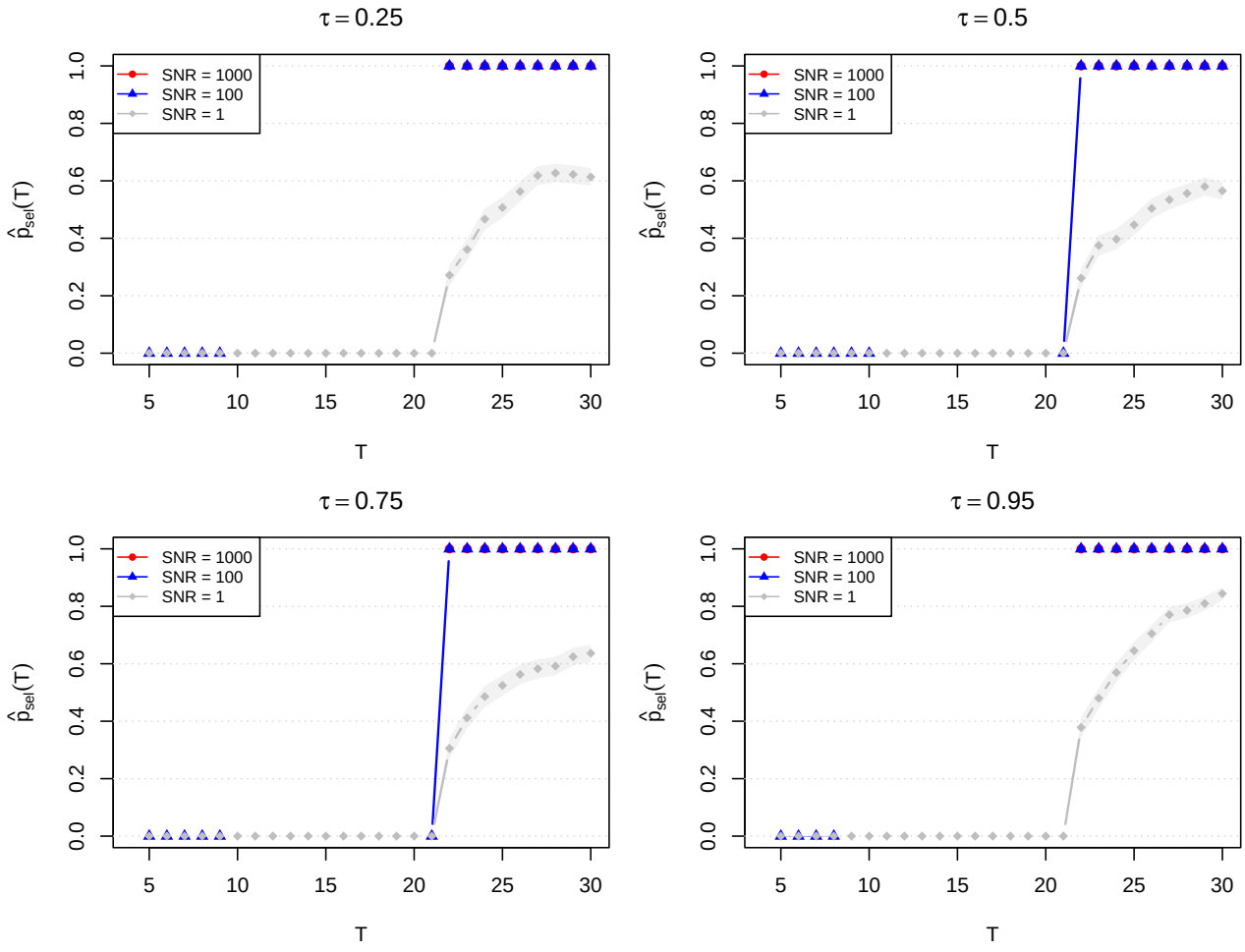


Figure 7: Empirical coverage of the selected operational intervals as a function of the observation horizon T, based on $M = 1000$ independent simulated trajectories for each combination of τ and SNR. Coverage is calculated conditionally on interval selection. The horizontal axis gives the observation horizon, and each panel corresponds to one quantile level.

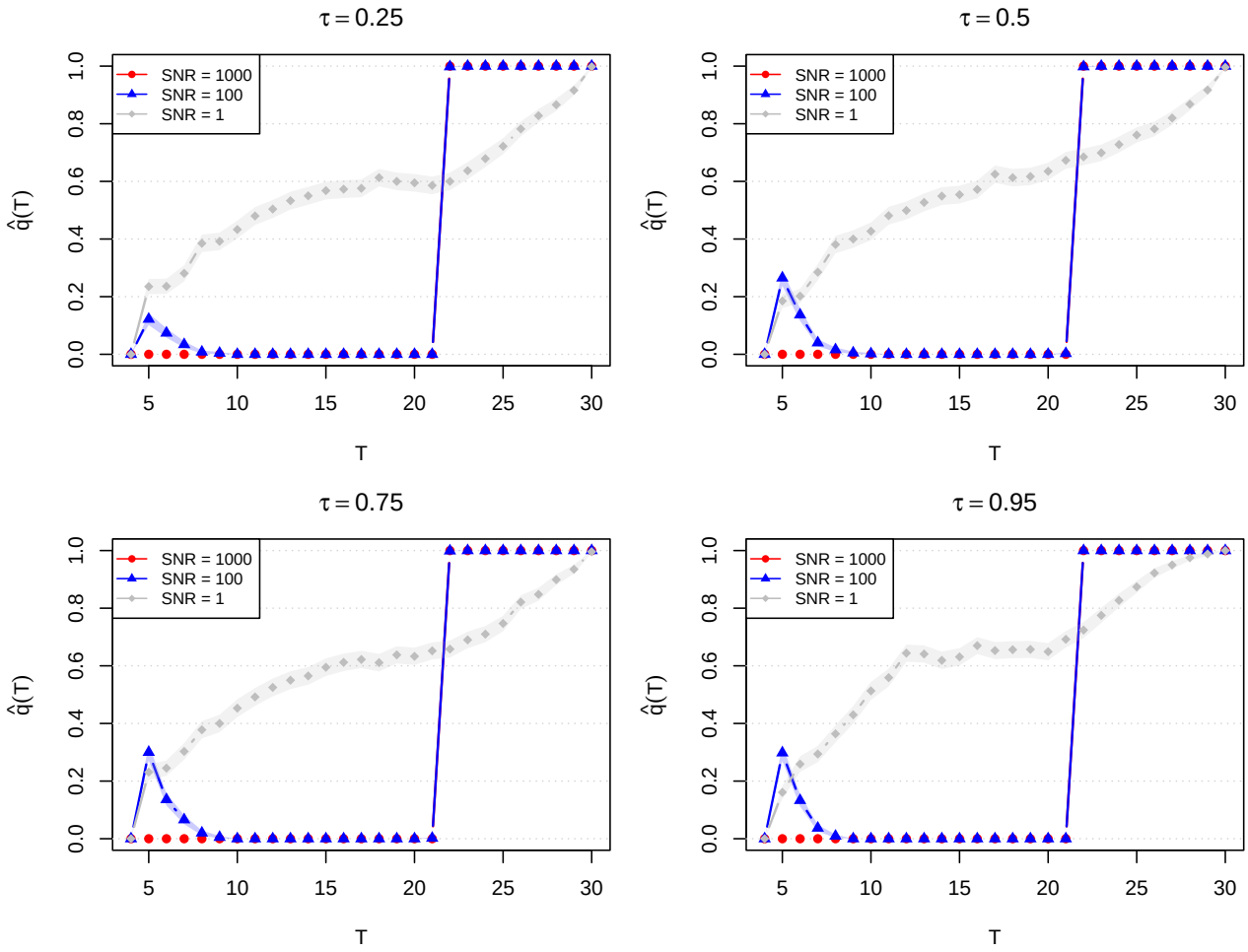


Figure 8: Empirical interval-selection rate as a function of the observation horizon T, based on $M = 1000$ independent simulated trajectories for each combination of τ and SNR. The selection rate is the proportion of replicates for which the operational rule selects an interval. Each panel corresponds to one quantile level.

8 Real-World Application: COVID-19 Case Study

We now demonstrate the application of our quantile-based peak estimation framework to real-world data. We focus on the daily confirmed COVID-19 cases in China, France, and Germany. Figure 9 displays the daily case series for these three countries. The Chinese series is obtained from the Johns Hopkins University Center for Systems Science and Engineering (JHU CSSE) feed via the `COVID19` R package (Guidotti and Ardia, 2020; Guidotti, 2022), while the French and German series are obtained from the European Centre for Disease Prevention and Control (ECDC) open data portal (ECDC, 2020). For each country, we analyze the sequence $T \mapsto \hat{l}(T)$ for the quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, using quantile-loss minimization over the Gaussian-shaped parametric peak model. Following Algorithm 1, the candidate peak location ranges over the analysis horizon $h = \max(T)$, so that for each country the search set is $\{1, \ldots, h\}$. We then assess the temporal stabilization of the estimated peak-location path using the retrospective stabilization criterion introduced in Section 6, applied separately to each country and quantile level; the resulting stabilization time is denoted T_{stab}. For each country we additionally apply the operational two-stage entry–exit construction of Section 7, which uses the oscillations of the estimated peak-location path to identify a candidate peak-region window $(T^{(k)}_{\text{entry}}, T^{(k)}_{\text{exit}})$ and whose width is denoted $w_k = T^{(k)}_{\text{exit}} - T^{(k)}_{\text{entry}}$. Because the observed peak location L is known for each of the three country-level trajectories used here, the resulting Stage-2 candidates can be evaluated retrospectively, and the narrowest candidate satisfying the strict containment criterion $T^{(k)}_{\text{entry}} < L < T^{(k)}_{\text{exit}}$ is reported for each country and quantile level, when such a candidate exists. This minimum-width selection is used only for retrospective evaluation and is not part of the operational procedure when L is unknown.

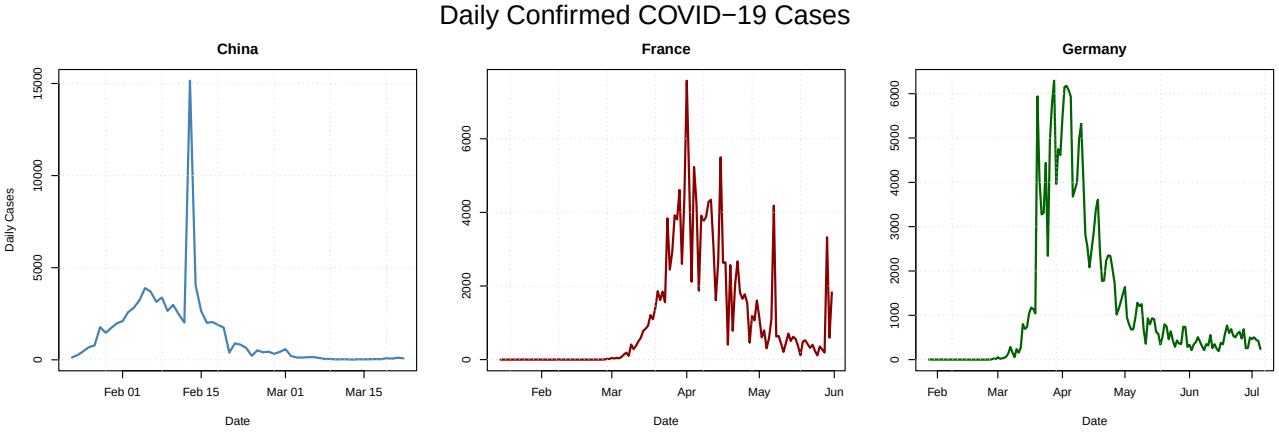


Figure 9: Daily confirmed COVID-19 cases in China, France, and Germany (left to right).

8.1 China

We analyze the first wave of the COVID-19 epidemic in China, using the first 60 daily new case counts from January 22 to March 22, 2020, obtained from the JHU CSSE feed via the `COVID19` R package (Guidotti and Ardia, 2020; Guidotti, 2022). The estimation is performed for observation horizons $T = 4, \ldots, 35$. For each quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we

compute the sequence $T \mapsto \hat{l}(T)$ by minimizing the empirical quantile loss over the Gaussian-shaped peak model, with candidate peak locations ranging over $l \in \{1, \ldots, h\}$ with $h = 35$. Temporal stabilization is assessed using the criterion introduced in Section 6, with tolerance $\varepsilon = 3$. The resulting estimated peak-location sequences are reported in Table **??**.

The corresponding stabilization times are $T_{\text{stab}} = 15$, 13, 23, and 28 for $\tau = 0.25$, 0.5, 0.75, and 0.95, respectively. The stabilization criterion characterizes the temporal behavior of the estimated peak-location path and is distinct from the entry–exit construction introduced in Section 7. For the present real-data application, where the observed peak location is known retrospectively, we apply the latter construction and, among the Stage-2 candidates satisfying $T^{(k)}_{\text{entry}} < L < T^{(k)}_{\text{exit}}$, retain the candidate with the smallest width $w_k = T^{(k)}_{\text{exit}} - T^{(k)}_{\text{entry}}$. This minimum-width selection is used only for retrospective evaluation and is not part of the operational procedure when the peak location is unknown. The candidate index k is therefore chosen among the peak-containing candidates by this smallest-width criterion.

For China, the observed peak location is $L = 22$. The resulting retrospective selection for the four quantile levels is reported in Table 5. For $\tau = 0.25$, the selected candidate is indexed by $k = 1$. It has $\epsilon'_1 = 1$, $T^{(1)}_{\text{exit}} = 26$, $T^*_1 = 15$, $\epsilon_1 = 3$, and $T^{(1)}_{\text{entry}} = 15$, and therefore has width $w_1 = 11$. For $\tau = 0.5$, the selected candidate is indexed by $k = 2$. It has $\epsilon'_2 = 2$, $T^{(2)}_{\text{exit}} = 24$, $T^*_2 = 15$, $\epsilon_2 = 2$, and $T^{(2)}_{\text{entry}} = 15$, and therefore has width $w_2 = 9$. For $\tau = 0.95$, the selected candidate is indexed by $k = 4$. It has $\epsilon'_4 = 15$, $T^{(4)}_{\text{exit}} = 26$, $T^*_4 = 21$, $\epsilon_4 = 21$, and $T^{(4)}_{\text{entry}} = 13$, and therefore has width $w_4 = 13$. For $\tau = 0.25$ and $\tau = 0.5$, the Stage-2 level ϵ_k is attained at a single observation horizon, so that $T^*_k = T^{(k)}_{\text{entry}}$. For $\tau = 0.95$, the level is attained at nine consecutive observation horizons spanning $T = 13$ to $T = 21$, so that $T^*_4 = 21$ while $T^{(4)}_{\text{entry}} = 13$. For $\tau = 0.75$, no Stage-2 candidate satisfies the strict containment condition $T^{(k)}_{\text{entry}} < 22 < T^{(k)}_{\text{exit}}$, and consequently no retrospective candidate is selected.

Table 5: Retrospective selection of the peak-region candidate for COVID-19 in China.

τ	k	ϵ'_k	$T^{(k)}_{\text{exit}}$	T^*_k	ϵ_k	$T^{(k)}_{\text{entry}}$	w_k	L
0.25	1	1	26	15	3	15	11	22
0.50	2	2	24	15	2	15	9	22
0.75	–	–	–	–	–	–	–	22
0.95	4	15	26	21	21	13	13	22

For $\tau = 0.5$, the selected window is $(15, 24)$. This is close to the window $(14, 25)$ reported in Dermoune et al. (2026) for the same dataset under the least-absolute-deviation criterion, the small difference reflecting that the peak-location trajectory is here recomputed for the quantile-based procedure.

The results show that the retrospective peak-region candidate depends on the quantile level. The selected windows are $(15, 26)$ for $\tau = 0.25$, $(15, 24)$ for $\tau = 0.5$, and $(13, 26)$ for $\tau = 0.95$. For $\tau = 0.75$, the two-stage construction does not produce a candidate whose window strictly contains the observed peak location $L = 22$. These results complement the stabilization analysis in Table **??**: the stabilization time describes the temporal stabilization of the estimated peak-

location path, whereas the two-stage construction uses its oscillations to identify candidate windows for the peak region.

8.2 France

Next, we apply the estimation method to daily COVID-19 cases in France, using ECDC data from January 15 to May 31, 2020 (ECDC, 2020). The analysis covers observation horizons $T = 50, \ldots, 89$. For each quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute the corresponding estimation path $T \mapsto \hat{l}(T)$, with candidate peak locations ranging over $l \in \{1, \ldots, h\}$ with $h = \max(T) = 89$. Temporal stabilization is assessed using the criterion introduced in Section 6, with tolerance $\varepsilon = 5$. The estimated peak-location sequences are reported in Table **??**.

The corresponding stabilization times are $T_{\text{stab}} = 85$, 82, 80, and 85 for $\tau = 0.25$, 0.5, 0.75, and 0.95, respectively. As in the China analysis, we then apply the two-stage entry–exit construction of Section 7 and select the peak-containing candidate with the smallest width, using the same minimum-width criterion introduced above.

For France, the observed peak location is $L = 78$. The resulting retrospective selection for the four quantile levels is reported in Table 6. For $\tau = 0.25$, the selected candidate is indexed by $k = 5$. It has $\epsilon'_5 = 13$, $T^{(5)}_{\text{exit}} = 79$, $T^*_5 = 76$, $\epsilon_5 = 13$, and $T^{(5)}_{\text{entry}} = 72$, and therefore has width $w_5 = 7$. For $\tau = 0.5$, the selected candidate is indexed by $k = 3$. It has $\epsilon'_3 = 4$, $T^{(3)}_{\text{exit}} = 82$, $T^*_3 = 78$, $\epsilon_3 = 6$, and $T^{(3)}_{\text{entry}} = 73$, and therefore has width $w_3 = 9$. For $\tau = 0.75$, the selected candidate is also indexed by $k = 3$. It has $\epsilon'_3 = 8$, $T^{(3)}_{\text{exit}} = 79$, $T^*_3 = 72$, $\epsilon_3 = 11$, and $T^{(3)}_{\text{entry}} = 70$, and therefore has width $w_3 = 9$. For $\tau = 0.95$, the selected candidate is indexed by $k = 6$. It has $\epsilon'_6 = 11$, $T^{(6)}_{\text{exit}} = 80$, $T^*_6 = 77$, $\epsilon_6 = 13$, and $T^{(6)}_{\text{entry}} = 70$, and therefore has width $w_6 = 10$. Unlike the China case, the plateau at level ϵ_k is not a singleton in any of the four selections: the sets $Z_k = \{T : \omega_l(T, T^{(k)}_{\text{exit}}) = \epsilon_k\}$ contain 5, 6, 3, and 8 observation times, respectively, so that $T^*_k > T^{(k)}_{\text{entry}}$ in every case.

Table 6: Retrospective selection of the peak-region candidate for COVID-19 in France.

τ	k	ϵ'_k	$T^{(k)}_{\text{exit}}$	T^*_k	ϵ_k	$T^{(k)}_{\text{entry}}$	w_k	L
0.25	5	13	79	76	13	72	7	78
0.50	3	4	82	78	6	73	9	78
0.75	3	8	79	72	11	70	9	78
0.95	6	11	80	77	13	70	10	78

The results show that the retrospective peak-region candidate depends on the quantile level. The selected windows are $(72, 79)$ for $\tau = 0.25$, $(73, 82)$ for $\tau = 0.5$, $(70, 79)$ for $\tau = 0.75$, and $(70, 80)$ for $\tau = 0.95$. All four windows strictly contain the observed peak location $L = 78$, so a retrospective candidate is selected for every quantile level, and the widths vary between 7 and 10 across the four quantile specifications.

8.3 Germany

Finally, we consider COVID-19 daily cases in Germany from January 28 to July 5, 2020, obtained from the ECDC open data portal (ECDC, 2020). To align the peak location with day $t = 61$, we index the data starting on January 28. We analyze observation horizons $T = 40, \ldots, 75$. For each quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute the estimation path $T \mapsto \hat{l}(T)$, with candidate peak locations ranging over $l \in \{1, \ldots, h\}$ with $h = 75$. Temporal stabilization is assessed using the criterion introduced in Section 6, with tolerance $\varepsilon = 15$. The resulting estimation paths are reported in Table **??**.

The corresponding stabilization times are $T_{\text{stab}} = 60$, 59, 56, and 57 for $\tau = 0.25$, 0.5, 0.75, and 0.95, respectively. As in the China and France analyses, we then apply the two-stage entry–exit construction of Section 7 and select the peak-containing candidate with the smallest width, using the same minimum-width criterion introduced above.

For Germany, the observed peak location is $L = 61$. The resulting retrospective selection for the four quantile levels is reported in Table 7. For $\tau = 0.25$, the selected candidate is indexed by $k = 7$. It has $\epsilon'_7 = 15$, $T^{(7)}_{\text{exit}} = 62$, $T^*_7 = 60$, $\epsilon_7 = 15$, and $T^{(7)}_{\text{entry}} = 60$, and therefore has width $w_7 = 2$. For $\tau = 0.95$, the selected candidate is indexed by $k = 6$. It has $\epsilon'_6 = 9$, $T^{(6)}_{\text{exit}} = 63$, $T^*_6 = 56$, $\epsilon_6 = 18$, and $T^{(6)}_{\text{entry}} = 56$, and therefore has width $w_6 = 7$. In both cases, the level ϵ_k is attained at a single observation horizon, so that $T^*_k = T^{(k)}_{\text{entry}}$. For $\tau = 0.5$ and $\tau = 0.75$, no Stage-2 candidate satisfies the strict containment condition $T^{(k)}_{\text{entry}} < 61 < T^{(k)}_{\text{exit}}$, and consequently no retrospective candidate is selected for these two quantile levels.

Table 7: Retrospective selection of the peak-region candidate for COVID-19 in Germany.

τ	k	ϵ'_k	$T^{(k)}_{\text{exit}}$	T^*_k	ϵ_k	$T^{(k)}_{\text{entry}}$	w_k	L
0.25	7	15	62	60	15	60	2	61
0.50	–	–	–	–	–	–	–	61
0.75	–	–	–	–	–	–	–	61
0.95	6	9	63	56	18	56	7	61

The results show that the retrospective peak-region candidate depends on the quantile level. The selected windows are $(60, 62)$ for $\tau = 0.25$, and $(56, 63)$ for $\tau = 0.95$. For $\tau = 0.5$ and $\tau = 0.75$, the two-stage construction does not produce a candidate whose window strictly contains the observed peak location $L = 61$. These results complement the stabilization analysis in Table **??**: the stabilization time describes the temporal stabilization of the estimated peak-location path, whereas the two-stage construction uses its oscillations to identify candidate windows for the peak region.

8.4 Comparison Across Quantile Levels

To compare the four quantile specifications, we evaluate the minimized quantile-loss objective at the final observation horizon $T_{\max}$ for each country. This provides a descriptive comparison of the empirical objective values obtained under the different quantile levels considered. The

quantile level yielding the smallest objective value at $T_{\max}$ is reported for each data set. The resulting minima are obtained for

- China: $\tau = 0.25$ at $T = 35$,
- France: $\tau = 0.95$ at $T = 89$,
- Germany: $\tau = 0.95$ at $T = 75$.

These results indicate that the empirical quantile-loss criterion favors different quantile specifications across the three epidemic trajectories. In particular, the lower quantile $\tau = 0.25$ yields the smallest objective value for China, whereas the upper quantile $\tau = 0.95$ yields the smallest objective value for both France and Germany. This comparison is descriptive and is intended to characterize the behavior of the proposed estimator under the four quantile levels considered, rather than to establish that any one of these quantiles represents the true conditional quantile of the epidemic process.

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Data Availability Statement The daily COVID-19 case counts analyzed in Section 8 are derived from two publicly available sources. The Chinese series is obtained from the Johns Hopkins University Center for Systems Science and Engineering (JHU CSSE) feed through the `COVID19` R package, covering the period 22 January to 22 March 2020. The French and German series are obtained from the European Centre for Disease Prevention and Control (ECDC) open data portal, covering the periods 15 January to 31 May 2020 and 28 January to 5 July 2020, respectively. All data used in this study are publicly available at the sources cited above and in the bibliography.

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